Doctor Appointment Booking Demo using n8n

Step 1 — Create a New Workflow

1. Login to n8n.
2. Click **New Workflow**.
3. Rename it as

```
Doctor Appointment Booking Demo
```

Save the workflow.

Step 2 — Add Form Trigger

Search for

```
Form Trigger
```

Configure:

Form Title

```
Doctor Appointment Request
```

Description

```
Select an available time slot to request a doctor appointment.
```

Add Form Fields

Patient Name

Property	**Value**
Type	Text
Required	Yes
Placeholder	Amal Perera

Patient Email

Property	**Value**
Type	Email

Property	Value
Required	Yes

Appointment Date

Property	Value
Type	Date
Required	Yes

Time Slot

Type:

```
Dropdown
```

Options:

```
09:00-09:30

09:30-10:00

10:00-10:30

10:30-11:00

14:00-14:30

14:30-15:00
```

Required:

```
Yes
```

Appointment Type

Dropdown options:

```
General Consultation

Follow-up Appointment

Prescription Discussion

Routine Check-up
```

Short Note

Type:

```
Textarea
```

Placeholder:

```
General appointment demonstration.
```

Required:

```
Yes
```

Form Options

Button Label

```
Submit Appointment Request
```

Path

```
doctor-appointment-demo
```

Run the node once to preview the form.

Step 3 — Prepare Appointment Data

Add a

```
Edit Fields
```

node.

Rename it

```
Prepare Appointment Data
```

Enable

```
Include Other Input Fields
```

Add Field 1

Name

```
startDateTime
```

Expression

```
{{ DateTime.fromISO($json.appointmentDate + 'T' + $json.timeSlot.split('-')[0],
```

```
{ zone: 'Asia/Colombo' }).toISO() }}
```

Add Field 2

Name

```
endDateTime
```

Expression

```
{{ DateTime.fromISO($json.appointmentDate + 'T' + $json.timeSlot.split('-')[1],
{ zone: 'Asia/Colombo' }).toISO() }}
```

Why is this node needed?

The form provides:

```
Date

2026-07-20

Time

09:00-09:30
```

Google Calendar requires:

```
2026-07-20T09:00:00+05:30

2026-07-20T09:30:00+05:30
```

This node prepares those timestamps.

Step 4 — Google Calendar Availability Check

Add

```
Google Calendar
```

Rename it

```
Check Doctor Availability
```

Operation:

```
Calendar Availability
```

Authenticate your Google account.

Choose your appointment calendar.

Configure:

Time Min

```
{{$json.startDateTime}}
```

Time Max

```
{{$json.endDateTime}}
```

Timezone

```
Asia/Colombo
```

Output Format

```
Availability
```

This node returns

```
available = true
```

or

```
available = false
```

Step 5 — IF Node

Add

```
IF
```

Rename it

```
Is Slot Available?
```

Condition

Left Value

```
{{$json.available}}
```

Operation

```
Boolean

is true
```

Workflow branches:

True

```
Appointment is available
```

False

```
Appointment already exists
```

Step 6 — FALSE Branch (Unavailable)

From the **False** output of the IF node, add a **Form** node.

Rename it:

```
Unavailable Page
```

Operation

```
Completion Page
```

Title

```
Time Slot Unavailable
```

Completion Message

```
{{ 'The doctor is already booked during the selected time slot.<br><br>' +
'<b>Date:</b> ' + $('Prepare Appointment Data').item.json.appointmentDate +
'<br><b>Time:</b> ' + $('Prepare Appointment Data').item.json.timeSlot +
'<br><br>Please submit another request using a different time slot.' }}
```

The workflow ends here for unavailable appointments.

Step 7 — TRUE Branch (Create Appointment)

From the **True** output of the IF node, add another

```
Google Calendar
```

Rename it

```
Create Doctor Appointment
```

Operation

```
Create Event
```

Start

```
{{$('Prepare Appointment Data').item.json.startDateTime}}
```

End

```
{{$('Prepare Appointment Data').item.json.endDateTime}}
```

Summary

```
{{'Doctor Appointment - ' + $('Prepare Appointment
Data').item.json.patientName}}
```

Location

```
Doctor Appointment Demo
```

Description

```
{{'Patient: ' + $('Prepare Appointment Data').item.json.patientName +
'\nEmail: ' + $('Prepare Appointment Data').item.json.patientEmail +
'\nAppointment type: ' + $('Prepare Appointment Data').item.json.appointmentType
+
'\nShort note: ' + $('Prepare Appointment Data').item.json.shortNote}}
```

Show Me As

```
Busy
```

Step 8 — Google Gemini Chat Model

Add

```
Google Gemini Chat Model
```

Authenticate with your Gemini API key.

Leave the default settings unless you want to customize the model.

Step 9 — AI Email Generator

Add

```
Basic LLM Chain
```

Rename it

```
Generate AI Confirmation
```

Connect the Gemini Chat Model to the AI Language Model port.

Use this prompt:

```
Write a short professional doctor appointment confirmation email.

Use only the supplied appointment information.
Do not provide medical advice.
Do not invent clinic names, policies, phone numbers, contact details, or extra
services.
Do not include an email subject.
Keep the message under 90 words.
Return only the plain-text email body.

Patient: {{ $('Prepare Appointment Data').item.json.patientName }}
Appointment type: {{ $('Prepare Appointment Data').item.json.appointmentType }}
```

```
Date: {{ $('Prepare Appointment Data').item.json.appointmentDate }}
Time slot: {{ $('Prepare Appointment Data').item.json.timeSlot }}
Note: {{ $('Prepare Appointment Data').item.json.shortNote }}

End with exactly:
This is an automatically generated confirmation from the doctor appointment
demo.
```

Step 10 — Send Email

Add

```
Gmail
```

Rename it

```
Send a Message
```

Authenticate Gmail.

To

```
{{$('Prepare Appointment Data').item.json.patientEmail}}
```

Subject

```
Doctor Appointment Confirmed
```

Email Type

```
Text
```

Message

```
{{$('Generate AI Confirmation').item.json.text}}
```

Step 11 — Success Page

Add a

```
Form
```

Rename it

```
Success Page
```

Operation

```
Completion Page
```

Title

```
Success Page
```

Completion Message

```
{{ 'Your doctor appointment request was completed successfully.<br><br>' +
'<b>Patient:</b> ' + $('Prepare Appointment Data').item.json.patientName +
'<br><b>Appointment type:</b> ' + $('Prepare Appointment
Data').item.json.appointmentType +
'<br><b>Date:</b> ' + $('Prepare Appointment Data').item.json.appointmentDate +
'<br><b>Time:</b> ' + $('Prepare Appointment Data').item.json.timeSlot +
'<br><br>A confirmation email has been sent to ' +
$('Prepare Appointment Data').item.json.patientEmail + '.' }}
```

Step 12 — Test the Workflow

1. Save the workflow.
2. Click **Test Workflow**.
3. Open the generated Form URL.
4. Submit an appointment.
5. Observe the workflow execution.
6. Check:
    - Google Calendar for the created event.
    - Gmail for the confirmation email.
    - Browser for the success page.

Try submitting the same date and time again to verify that the **Unavailable Page** is displayed.

Workflow Summary

Step	Node
1	Form Trigger
2	Set (Prepare Appointment Data)
3	Google Calendar (Availability Check)
4	IF (Available?)
5	Form (Unavailable Page)
6	Google Calendar (Create Event)
7	Google Gemini Chat Model
8	Basic LLM Chain (Generate AI Confirmation)
9	Gmail (Send Email)
10	Form (Success Page)

Learning Outcomes

After completing this practical, students will be able to:

- Build an end-to-end workflow using n8n.
- Design interactive forms with the Form Trigger node.
- Use the Set node to transform data.
- Integrate Google Calendar for scheduling.
- Implement decision-making with the IF node.
- Generate dynamic content using Google Gemini AI.
- Send automated emails using Gmail.
- Create user-friendly success and error pages.
- Test and debug workflow executions.